

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
krvalieva@krvalieva:~$ ls /etc/ > file.txt
krvalieva@krvalieva:~$ ls >> file.txt
krvalieva@krvalieva:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
```

Рисунок 1: Запись в файл

```
krvalieva@krvalieva:~$  
krvalieva@krvalieva:~$ grep .conf file.txt > conf.txt  
krvalieva@krvalieva:~$ cat conf.txt  
anthy-unicode.conf  
asound.conf  
brltty.conf  
chrony.conf  
cpupower-service.conf  
dconf  
dnsmasq.conf  
dracut.conf  
dracut.conf.d  
fprintd.conf  
fuse.conf  
host.conf  
idmapd.conf  
kdump.conf  
krb5.conf  
krb5.conf.d  
ld.so.conf  
ld.so.conf.d  
libaudit.conf  
libuser.conf
```

Рисунок 2: Поиск расширения .conf


```
00000000
/home/krvalieva/.npm/_cacache/index-v5/f3/6c/c3986891a2eed646eaa7d914d91cffc8d6421272c959bdb5a8748
6645945
/home/krvalieva/.npm/_cacache/index-v5/84/ce
/home/krvalieva/.npm/_cacache/index-v5/ce
/home/krvalieva/.npm/_cacache/index-v5/09/c0
/home/krvalieva/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ced5cf6fc29def3022
0242c53
/home/krvalieva/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58996cfad55fb2000f5
9e2edcc
/home/krvalieva/.npm/_cacache/index-v5/cd
/home/krvalieva/.npm/_cacache/index-v5/c9
/home/krvalieva/.npm/_cacache/index-v5/c0
/home/krvalieva/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b53cea310cda0b203
217f693
/home/krvalieva/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d1350482a1d99
0504757
/home/krvalieva/.npm/_cacache/index-v5/af/cf
/home/krvalieva/.npm/_cacache/index-v5/c3
/home/krvalieva/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f2506214f94db88
61160d3
/home/krvalieva/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c8188a44d554f
3f3ba2e
/home/krvalieva/.npm/_cacache/index-v5/ca
/home/krvalieva/snap/hugo/common
/home/krvalieva/snap/hugo/current
/home/krvalieva/conf.txt
krvalieva@krvalieva:~$
```

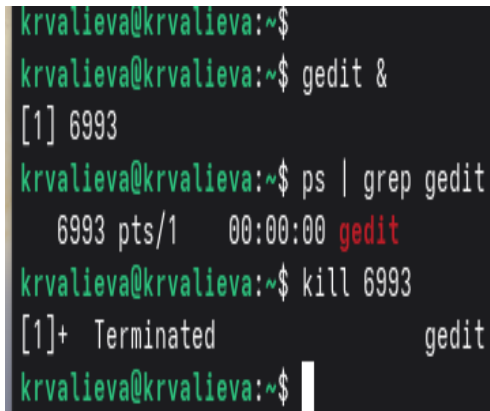
Рисунок 3: Поиск файлов

```
/etc/foomatic/hashe.s.d/hashe.s.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: '/etc/ssh/ssh_config.d': Permission denied
find: '/etc/sss.d': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
krvalieva@krvalieva:~$  
krvalieva@krvalieva:~$ find /etc -name "h*" -print | less  
find: '/etc/audit': Permission denied  
krvalieva@krvalieva:~$ find ~ -name "log*" > logfile &  
[1] 6961  
krvalieva@krvalieva:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
krvalieva@krvalieva:~$ rm logfile  
krvalieva@krvalieva:~$
```

Рисунок 5: Фоновый запуск процесса



```
krvalieva@krvalieva:~$  
krvalieva@krvalieva:~$ gedit &  
[1] 6993  
krvalieva@krvalieva:~$ ps | grep gedit  
    6993 pts/1    00:00:00 gedit  
krvalieva@krvalieva:~$ kill 6993  
[1]+  Terminated                  gedit  
krvalieva@krvalieva:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DE(1)                                User Commands                                DE(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space available on the file system containing each file name argument.  If no file name is given, the space available on all currently mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows the space available on that file system rather than on the file system containing the device node.  This version of df cannot show the space available on unmounted file systems, because on most kinds of systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
krvalieva@krvalieva:~$  
krvalieva@krvalieva:~$ df  
Filesystem      1K-blocks      Used Available Use% Mounted on  
/dev/nvme0n1p3 155186176 91083736 63495144 59% /  
devtmpfs         4011284         0    4011284  0% /dev  
tmpfs            4035656         100    4035556  1% /dev/shm  
tmpfs            1614264        1960    1612304  1% /run  
tmpfs             1024           0         1024  0% /run/credentials/systemd-journald.service  
tmpfs            4035660        1644    4034016  1% /tmp  
/dev/nvme0n1p3 155186176 91083736 63495144 59% /home  
/dev/nvme0n1p2 1992552    518800    1352512  28% /boot  
/dev/loop1       68480       68480         0 100% /var/lib/snapd/snap/core24/1349  
/dev/loop2       49280       49280         0 100% /var/lib/snapd/snap/snapd/25935  
/dev/loop0       115712     115712         0 100% /var/lib/snapd/snap/hugo/25676  
tmpfs             1024           0         1024  0% /run/credentials/systemd-resolved.service  
tmpfs            807128        176     806952  1% /run/user/1057  
/dev/sr0         2677920     2677920         0 100% /run/media/krvalieva/Fedora-WS-Live-43  
krvalieva@krvalieva:~$
```

Рисунок 9: Справка по команде du

```
4      ./npm/_cacache/index-v5/a7
4      ./npm/_cacache/index-v5/30/1a
4      ./npm/_cacache/index-v5/30
424    ./npm/_cacache/index-v5
115644 ./npm/_cacache
115708 ./npm
12     ./snap/hugo/25676/.config/go/telemetry/local
0      ./snap/hugo/25676/.config/go/telemetry/upload
12     ./snap/hugo/25676/.config/go/telemetry
12     ./snap/hugo/25676/.config/go
12     ./snap/hugo/25676/.config
12     ./snap/hugo/25676
0      ./snap/hugo/common
16     ./snap/hugo
16     ./snap
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
1451588 .
krvalieva@krvalieva:~$
```

Рисунок 10: Запуск команды du


```
/home/krvalieva/.npm/_cacache/index-v5/ca/3d
/home/krvalieva/.npm/_cacache/index-v5/85
/home/krvalieva/.npm/_cacache/index-v5/85/e2
/home/krvalieva/.npm/_cacache/index-v5/a7
/home/krvalieva/.npm/_cacache/index-v5/a7/81
/home/krvalieva/.npm/_cacache/index-v5/30
/home/krvalieva/.npm/_cacache/index-v5/30/1a
/home/krvalieva/snap
/home/krvalieva/snap/hugo
/home/krvalieva/snap/hugo/25676
/home/krvalieva/snap/hugo/25676/.config
/home/krvalieva/snap/hugo/25676/.config/go
/home/krvalieva/snap/hugo/25676/.config/go/telemetry
/home/krvalieva/snap/hugo/25676/.config/go/telemetry/local
/home/krvalieva/snap/hugo/25676/.config/go/telemetry/upload
/home/krvalieva/snap/hugo/common
/home/krvalieva/monthly
/home/krvalieva/reports
/home/krvalieva/reports/monthly
/home/krvalieva/reports/monthly/monthly
/home/krvalieva/ski.places
/home/krvalieva/ski.places/equipment
/home/krvalieva/ski.places/plans
/home/krvalieva/australia
/home/krvalieva/play
/home/krvalieva/play/games
/home/krvalieva/play/games/play
krvalieva@krvalieva:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.